

Vibration and Damping Analysis of a Bridge Stay Cable

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Abstract—This report presents a 2D dynamic model of a stay cable represented as lumped masses with axial and bending elements. Gravity, initial tension, aerodynamic drag and lift, and mechanical damping are included. Using an implicit Newton-Raphson time integrator, the model captures cable vibrations under steady wind and shows stresses increasing toward the anchorage, peaking near the fixed end.

I. PROBLEM STATEMENT AND MOTIVATION

Stay cables are key load-carrying components of modern cable-stayed bridges. Because of their long, slender, and low inherent damping, they are prone to a variety of wind and rain-induced vibration mechanisms, including rain-wind-induced vibration and parametric excitation driven by deck motion [1]. These vibrations can lead to many problems, like fatigue damage and local component failure, if not properly controlled.

To better understand the interaction between wind loading, gravity, cable tension, and damping, this project develops a simplified numerical model of a single stay cable. The current version of the code focuses on a two-dimensional plane model. The cable is modeled as a sequence of lumped nodes connected by stretching and bending elements, and is subjected to gravity, steady uniform wind, aerodynamic drag and lift, and local mechanical damping.

The motivation for this work is twofold. The first objective is to establish a computational framework capable of resolving the time-varying dynamic response of a stay cable under realistic wind and damping conditions through a robust implicit time-integration scheme suited for stiff cable systems. The second objective is to provide a foundation for future extensions, including time-dependent boundary conditions such as deck vibration, multiple damping devices, and comprehensive vibration analyses in both time and frequency domains. This midterm report summarizes the current model formulation, implementation details, preliminary simulation results, and planned developments.

II. PHYSICAL MODEL AND ASSUMPTIONS

The cable examined in this project represents a typical long-span stay cable of approximately 300 m in length [2]. It is assumed to lie within a vertical plane and to be anchored between a lower deck point and an upper tower point. For computational purposes, the cable is represented by a sequence of n_v lumped-mass nodes connected by $n_e = n_v - 1$ edges of equal reference length, each edge acting as an axial

stretching element. Bending stiffness is introduced through a discrete curvature formulation applied to neighboring triplets of nodes, consistent with standard slender-structure rod and beam models.

The cable's cross-section is taken as a solid circular steel section of radius r_0 , from which the axial stiffness EA and bending stiffness EI follow directly. The initial geometry is defined as a straight inclined line between the two supports, after which a static equilibrium problem—including gravity and axial elasticity—is solved to obtain a sagged configuration with nonzero static tension. This state is essential because it governs the natural frequencies and overall dynamic behavior. The model is restricted to in-plane motion; out-of-plane dynamics, torsional deformation, and cross-ties are not considered at this stage. Aerodynamic effects are represented using classical drag and lift formulations for circular cylinders in cross-flow, with constant coefficients and without resolving spatial flow variability or rain-wind interaction mechanisms.

Taken together, these modeling choices produce a two-dimensional, edge-by-edge dynamic system that remains computationally efficient while still capturing the essential interactions among gravity, axial tension, bending stiffness, and aerodynamic loading.

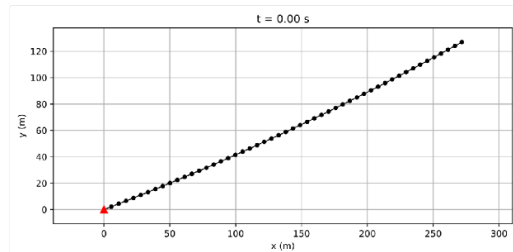


Fig. 1. Initial undeformed configuration of the discretized stay cable

III. NUMERICAL FORMULATION

The generalized coordinates are collected in a vector $\mathbf{q} \in \mathbb{R}^{2n_v}$, containing the x - and y -coordinates of each node:

$$\mathbf{q} = [x_0, y_0, x_1, y_1, \dots, x_{n_v-1}, y_{n_v-1}]^T.$$

Velocities are defined as $\mathbf{u} = \dot{\mathbf{q}}$. For node k , its position and velocity at time step n are denoted by $\mathbf{x}_k^n$ and $\mathbf{u}_k^n$, respectively.

A. Stretching energy

Each edge connects two adjacent nodes, denoted `node0` and `node1` in the code. The current edge vector is

$$\mathbf{e} = \text{node1} - \text{node0}, \quad L = \|\mathbf{e}\|,$$

where L is the current length, and l_k is the reference (initial) length of the same edge. The stretching strain is approximated as

$$\varepsilon_x = \frac{L}{l_k} - 1.$$

The stretching energy of the edge is

$$E_s = \frac{1}{2} EA l_k \varepsilon_x^2,$$

where EA is the axial stiffness (Young's modulus E times the cross-sectional area A). The code computes the gradient $\partial E_s / \partial \mathbf{q}$ and the Hessian $\partial^2 E_s / \partial \mathbf{q}^2$ analytically for each edge in 2D, and assembles them into the global stretching force vector $\mathbf{F}_s$ and Jacobian $\mathbf{J}_s$.

B. Bending energy

Bending resistance is introduced via a discrete curvature κ at interior nodes, based on three consecutive nodes—`node0`, `node1`, and `node2`. The local curvature is obtained from the turning angle between adjacent segments divided by an average segment length. The bending energy at an interior node is

$$E_b = \frac{1}{2} EI l_v (\kappa - \bar{\kappa})^2,$$

where EI is the bending stiffness, l_v is a Voronoi-type nodal length, and $\bar{\kappa}$ is the natural curvature (zero for a straight initial cable). The code evaluates the gradient and Hessian of E_b numerically using finite differences, yielding the bending force vector $\mathbf{F}_b$ and Jacobian $\mathbf{J}_b$ acting on the three involved nodes.

C. Mass matrix, gravity, and initial tension

Each node carries a lumped mass

$$m = \rho A \delta L,$$

where ρ is the material density, A is the cross-sectional area, and δL is the segment length. The global mass matrix $\mathbf{M}$ is diagonal, with identical mass entries for the x and y directions of each node.

Gravity is applied as a constant distributed load along the cable, leading to a nodal force vector $\mathbf{F}_g$ with vertical components proportional to the tributary length associated with each node.

To obtain a realistic initial configuration, a static equilibrium solve is performed under gravity and axial stiffness, with both ends fixed. The result is a sagged cable geometry and a consistent internal tension distribution that serves as the starting point for the dynamic simulation.

D. High-Reynolds-number drag

For each node k , the code computes a backward Euler estimate of the nodal velocity:

$$\mathbf{u}_k = \frac{\mathbf{x}_k^{n+1} - \mathbf{x}_k^n}{\Delta t},$$

where $\mathbf{x}_k^{n+1}$ and $\mathbf{x}_k^n$ are the new and old nodal positions, and Δt is the time step. The incoming wind velocity is represented by a constant vector $\mathbf{u}_f$ (for example, $\mathbf{u}_f = [U, 0]^T$ for horizontal wind of speed U).

The relative velocity between the cable and the flow at node k is then

$$\mathbf{u}_{\text{rel},k} = \mathbf{u}_k - \mathbf{u}_f.$$

The aerodynamic drag force on a 2D segment associated with node k is approximated as

$$\mathbf{F}_{\text{drag},k} = -\frac{1}{2} \rho_f C_D A_k \|\mathbf{u}_{\text{rel},k}\| \mathbf{u}_{\text{rel},k},$$

where ρ_f is the fluid density, C_D is the drag coefficient for a circular cylinder, and A_k is an effective projected area, taken as $2 r_0 l_{v,k}$ with r_0 the cable radius and $l_{v,k}$ the Voronoi length associated with node k . This expression follows the standard quadratic drag formula used for high-Reynolds-number cross-flow around circular cylinders. The code also derives the Jacobian of $\mathbf{F}_{\text{drag},k}$ with respect to $\mathbf{x}_k^{n+1}$ to construct the drag contribution to the global Jacobian $\mathbf{J}_h$.

E. Mechanical viscous damper

In addition to fluid-related damping, the model includes a viscous mechanical damper located at the mid-span node. This damper is represented by a diagonal damping matrix $\mathbf{C}$ that acts on the nodal velocities:

$$\mathbf{F}_v = -\mathbf{C} \mathbf{u},$$

where

$$\mathbf{u} = \frac{\mathbf{q}^{n+1} - \mathbf{q}^n}{\Delta t}$$

is the estimated velocity vector obtained using a backward Euler update. In the current implementation, only the vertical degree of freedom of the mid-span node is assigned a non-zero damping coefficient c , representing a vertical viscous damper anchored to the deck or tower. This term directly increases the effective damping of the cable's fundamental vertical mode.

F. Lift force model

A preliminary lift model is also implemented, analogous to the drag expression. For each node, the aerodynamic lift force is taken as

$$\mathbf{F}_{\text{lift},k} = -\frac{1}{2} \rho_f C_L A_k \|\mathbf{u}_{\text{rel},k}\|^2$$

where C_L is a lift coefficient, A_k is the same projected area used for drag, and the flow direction is to the incoming wind. By choosing appropriate C_L values based on aerodynamic data for stay cables, this term can be used to explore the influence of lift on vibration amplitudes and instability thresholds [3].

Altogether, the total non-gravitational external force on the cable at each time step is composed of the aerodynamic drag and lift, and the mechanical damper reaction. The dynamic response of stay cables is strongly governed by aerodynamic loading and structural damping. Together, these components enable the simulation framework to capture the essential fluid–structure interactions and energy dissipation pathways that dominate the vibration behavior of long-span cable systems.

IV. PROCESS AND PRELIMINARY RESULTS

The implementation is carried out in Python using NumPy for linear algebra and Matplotlib for visualization. The cable is discretized into $n_v = 51$ nodes, resulting in $n_e = 50$ edges. This choice of vertical elevation ($d_y \approx 127\text{m}$) falls within the typical range of stay-cable inclination angles, which are generally maintained above 25° in modern cable-stayed bridge designs [4]. Steel properties ($E \approx 200\text{GPa}$, $\rho \approx 7850\text{kg/m}^3$) and a representative cable radius are used to define EA , EI , and the linear mass density [2]. A constant gravitational field and a steady horizontal wind of prescribed speed are applied.

A vertical viscous damper is attached at the mid-span node via the damping matrix $\mathbf{C}$, with a specified damping coefficient c . The time step is chosen as $\Delta t = 0.01\text{s}$, and simulations are typically run for 10s of physical time, giving 1000 time steps. The code generates snapshots of the cable geometry every few time steps, which can be displayed interactively or saved as figures for later inspection.

Preliminary simulations indicate that the cable shape remains close to its initial sagged profile, as expected for a stay cable with significant initial tension. Small oscillations superposed on this mean shape are driven by the aerodynamic loading and are progressively damped by both fluid and mechanical damping mechanisms. When the axial stress along the cable is reconstructed from the stretching strain in each edge, it is observed that stress tends to increase from the mid-span toward the anchorage, with the maximum stress appearing near the fixed end. This behavior is consistent with the fact that the anchored end must carry the cumulative tension from the entire cable and is qualitatively in line with engineering observations of high-demand regions in stay cables.

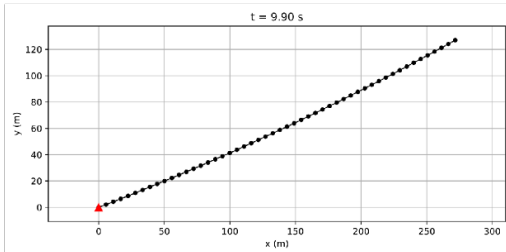


Fig. 2. Cable configuration after 10 seconds of aerodynamic and mechanical loading

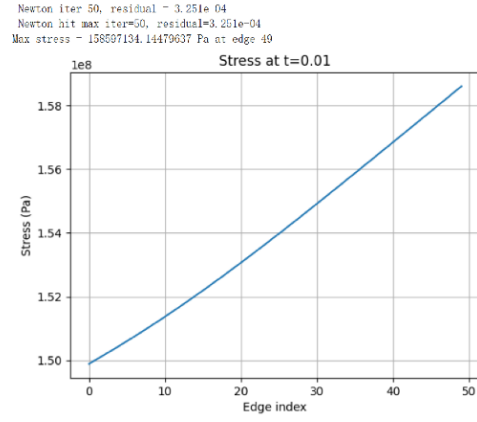


Fig. 3. Increasing tension toward the anchor

Current time: 2.6099999999999883

Fs (stretch): 2.528e+06 N
Fb (bend): 3.035e+02 N
Fg (gravity): 3.677e+04 N
Fh (drag): 2.792e+03 N
Fv (damping): 0.000e+00 N
Fl (lift): 8.377e+02 N

Fs (stretch): 2.528e+06 N
Fb (bend): 3.035e+02 N
Fg (gravity): 3.677e+04 N
Fh (drag): 2.791e+03 N
Fv (damping): 4.208e+02 N
Fl (lift): 8.372e+02 N

Fs (stretch): 2.528e+06 N
Fb (bend): 3.035e+02 N
Fg (gravity): 3.677e+04 N
Fh (drag): 2.791e+03 N
Fv (damping): 4.208e+02 N
Fl (lift): 8.372e+02 N

Fig. 4. Breakdown of total forces from stretching, bending, gravity, drag, damping, and lift at a specific time step

V. PLANNED EXTENSIONS AND VIBRATION ANALYSIS

Although the current model captures the time-varying motion of the cable under wind loading, gravity, and damping, several directions remain for further development. A first step will be to incorporate time-dependent boundary conditions, since the deck in practical bridges can undergo vibration due to traffic, wind, or seismic effects. Prescribing a moving boundary at the lower anchorage would introduce parametric excitation and allow examination of support-induced oscillations [1].

Another important extension involves enriching the damping representation. Real structures often employ multiple external dampers positioned at different locations along a stay cable, or use combined devices such as viscous dampers, tuned mass dampers, or inerter-based systems [5]. Expanding the damping matrix to accommodate such configurations would make the model more representative of engineering practice.

A further development will focus on systematic vibration analysis. While the current implementation primarily advances the solution in time, subsequent work will extract time histories of nodal displacements and stresses and analyze them using Fast Fourier Transform techniques [6]. This will enable identification of dominant frequencies, mode shapes, and potential resonances under varying wind speeds, damping arrangements, and boundary conditions. Together, these extensions will enhance the model's capability to investigate complex vibration mechanisms in stay cables and to assess the performance of different mitigation and control strategies.

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